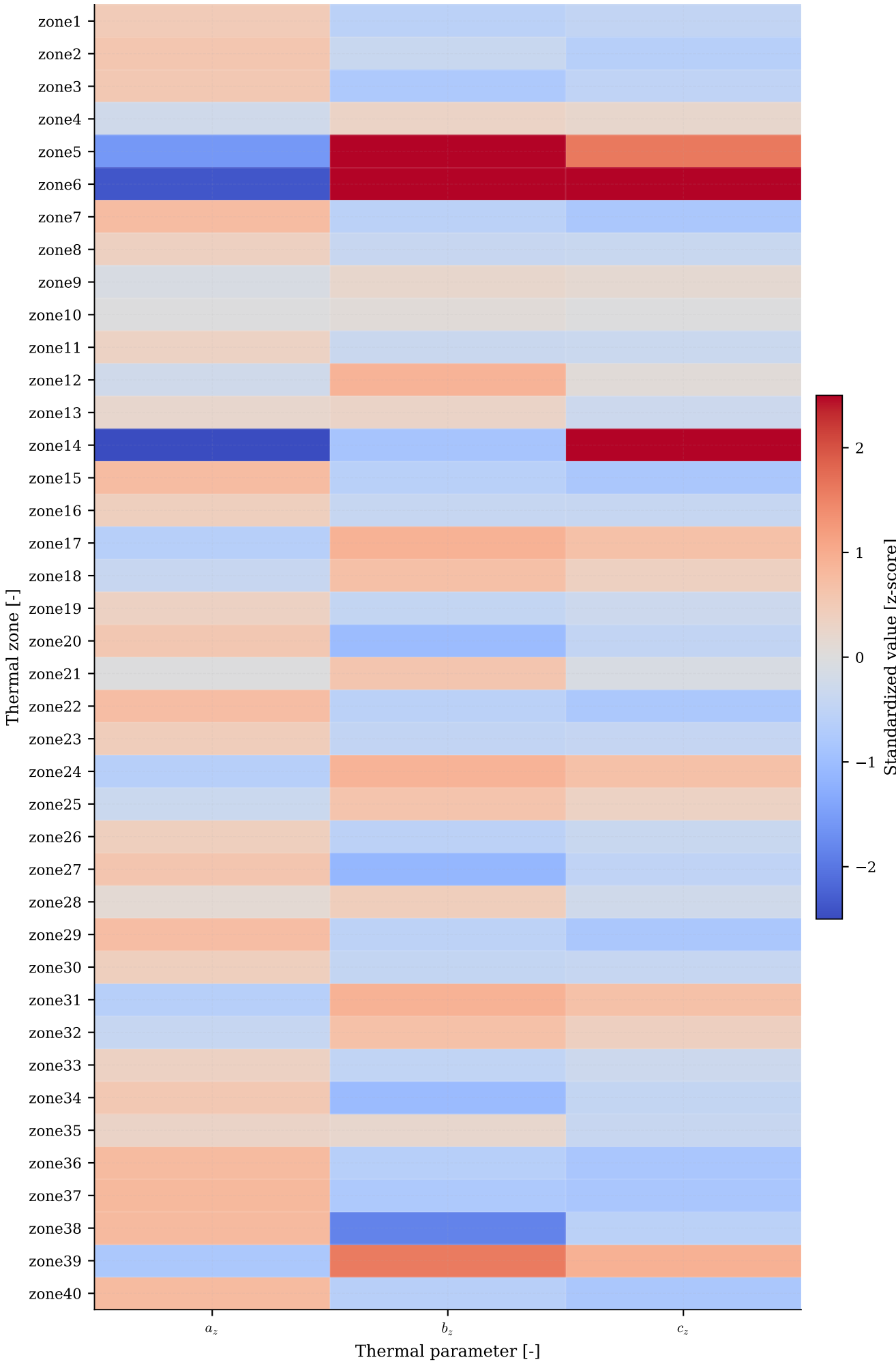


Normalized thermal-parameter patterns by zone



Column-wise z-score normalization: negative = below the cross-zone mean, positive = above the cross-zone mean.